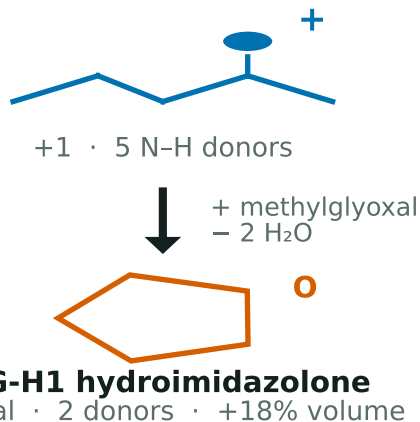


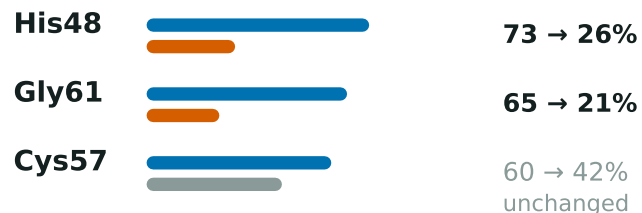
1 · THE MODIFICATION

Arg143 — catalytic arginine



2 · WHAT GLYCATION DOES

Site-143 core contact network



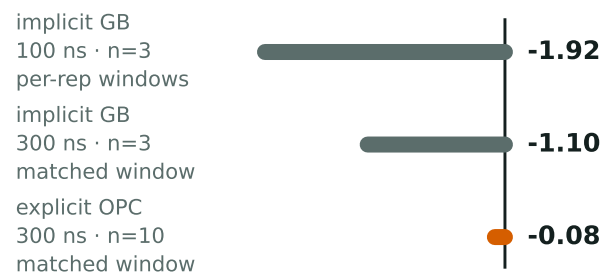
-36.1

percentage points, core composite
p = 0.0009 vs preregistered α = 0.0042
 n = 10 per group · explicit OPC water · 300 ns

3 · AND WHAT IT DEPENDS ON

Same endpoint, three methods

site H-bonds, glycated — native



The implicit-solvent result does not survive
 explicit water. Most AGE simulation work
 runs in implicit solvent.

Preregistered

endpoint and α sealed 22 Aug,
 before the data existed

Replicate-level

n = 10 per group; the replicate
 is the unit, never the frame

Window-robust

significant in all five analysis
 windows, including 0–300 ns

Fully reported

all six predictions, including
 two refuted and one unevaluable

Glycation at Arg143 breaks a loop-junction contact network without unfolding SOD1

Cys57 — the residue implicated through its disulfide with Cys146 — is the one contact that does not move.

Backbone RMSD, radius of gyration and β -sheet content are unchanged across twenty replicates: the perturbation is local, not global.